

## Lecture 6

# Multivariate Gaussians & Mixture of Gaussians

EECS 189/289, Spring 2026 @ UC Berkeley

### Assigned Book Chapters:

- **Probability:** Chapter 2.[1-2], 2.3.[1-3] (we will return to 2.4 onward later)
- **Distributions:** Chapter 3.1, 3.2.1 (the rest of 3 is more advanced Gaussians)
- **Clustering:** Chapter 15.1 (k-means), 15.2 (Gaussian Mixture Models)

# Roadmap

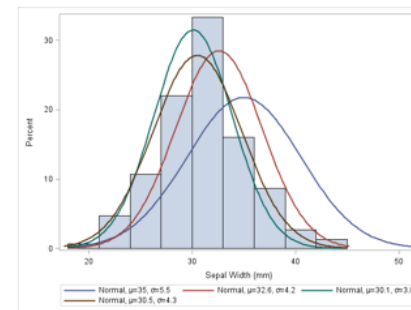
- Finish Maximum Likelihood Estimation (MLE)
- Multivariate Gaussians
- Mixture of Gaussians

# Roadmap

- Finish Maximum Likelihood Estimation (MLE)
- Multivariate Gaussians
- Mixture of Gaussians

# Recap from last class

## ML: main abstract ideas



- Training data set:  $D = \{(x_i, y_i)\}_{i=1}^N$   $x_i \in R^D$   
 $y_i \in R$  or  $y_i \in \{-1, 1\}$

$$D = \{(x_i)\}_{i=1}^N$$

- Model class:

aka hypothesis class

$$g(x | w, w_0) = w^T x + w_0$$

Linear Regression

$$p(x | \mu, \sigma) = N(x | \mu, \sigma)$$

Gaussian Probability Density

- Loss Function:  $L_i((x_i, y_i), \{w, w_i\}) = (y_i - g(x | w, w_0))^2$  “Squared Loss”

$$L_i(x_i, \{\mu, \sigma\}) = \log p(x_i; \mu, \sigma) = \log N(x_i | \mu, \sigma)$$

Log Likelihood loss

- Learning Objective:  $\operatorname{argmin}_{\phi} \sum_{i=1}^N L_i((x_i, y_i), \phi)$  Optimization Problem

# Recap from last class

## e.g. MLE for univariate Gaussian

- Find the setting of the parameters that set the partial derivatives to zero:

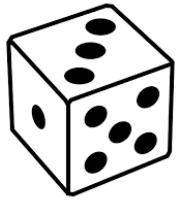
$$\mu_{MLE}, \sigma_{MLE}^2 = \underset{\theta \in \Theta}{\operatorname{argmax}} \log p(D|\theta) = \underset{\mu, \sigma^2}{\operatorname{argmax}} \sum_{i=1}^N \log p(x_i|\mu, \sigma^2)$$

- Lets expand out so we can take the derivative:

$$\begin{aligned} \frac{\partial}{\partial \mu} \sum_{i=1}^N \log p(x_i|\mu, \sigma^2) &= \sum_i \frac{\partial}{\partial \mu} \log \left[ \frac{1}{\sqrt{2\pi}\sigma^2} \exp\left(-\frac{1}{2\sigma^2}(x_i - \mu)^2\right) \right] \\ &= \sum_i \frac{\partial}{\partial \mu} \left[ \log(\sqrt{2\pi}\sigma^2) - \frac{1}{2\sigma^2}(x_i - \mu)^2 \right] \\ &= \sum_i \left[ 0 + \frac{1}{\sigma^2}(x_i - \mu) \right] \xrightarrow{\text{set to zero}} \sum_i x_i = \sum_i \mu \end{aligned}$$

$\sum_i x_i = N\mu$   
 $\Rightarrow \mu = \frac{\sum_i x_i}{N}$

# e.g. MLE for the multinomial distribution



- Consider a six-sided die that we will roll: we want to know the probability of each side of the die turning up ( $\theta = \theta_1 \dots \theta_6$ ).

- Assume

- We write

- Lets use

$$X \sim N(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp - \frac{(x - \mu)^2}{2\sigma^2}$$

- First, since one side must always face up, we know that  $1 = \sum_k \theta_k$ .

- Now we write the likelihood:

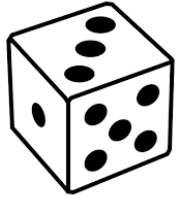
$$P(D|\theta) = p(x_1, \dots, x_N|\theta) = \prod_{i=1}^N p(x_i|\theta) = \prod_{i=1}^N \prod_{k=1}^6 \theta_k^{I[x_i=k]} = \prod_{k=1}^6 \theta_k^{\sum_i I[x_i=k]} = \prod_{k=1}^6 \theta_k^{n_k}$$

Now our MLE problem becomes:

$$\theta_{MLE} = \operatorname{argmax}_{\theta \in \Theta} \log p(D|\theta) = \operatorname{argmax}_{\theta \in \{\Theta | 1 = \sum_k \theta_k\}} \sum_{k=1}^6 \log \theta_k^{n_k}$$

constrained optimization

e.g. MLE for the multinomial distribution



Have a constrained optimization problem:

$$\theta_{MLE} = \underset{\theta \in \Theta}{\operatorname{argmax}} \log p(D|\theta) = \underset{\theta \in \{\Theta | 1 = \sum_k \theta_k\}}{\operatorname{argmax}} \sum_{k=1}^6 \log \theta_k^{n_k}$$

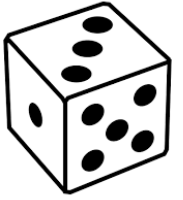
*Handwritten notes:*  $n_k = |\{i | x_i = k\}|$  (in blue),  $n_k$  (circled in blue), and *constrained optimization* (in green).

What is one technique you should have learned in first year calculus to solve this?

The technique of [Lagrange multipliers](#) (Appendix C of textbook):

$$J(\theta, \lambda) = \log p(D|\theta) + \lambda(1 - \sum_k \theta_k) \text{ (look for stationary points wrt } \theta, \lambda)$$

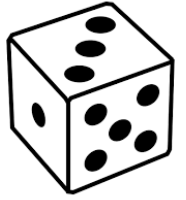
e.g. MLE for the multinomial distribution



$$J(\theta, \lambda) = \log p(D|\theta) + \lambda(1 - \sum_k \theta_k) = \sum_{k=1}^6 \log \theta_k^{n_k} + \lambda(1 - \sum_k \theta_k)$$

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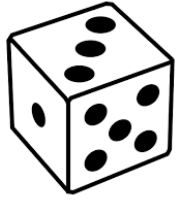
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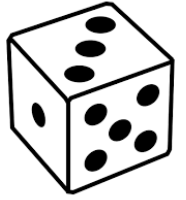
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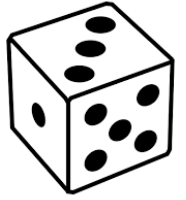


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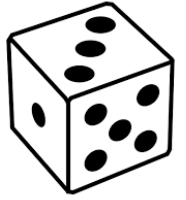


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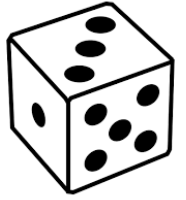
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3. Lets plug this into 1),  $1 = \sum_k \theta_k = \sum_k \frac{n_k}{\lambda} \Rightarrow \lambda = \sum_k n_k = N$ .

e.g. MLE for the multinomial distribution



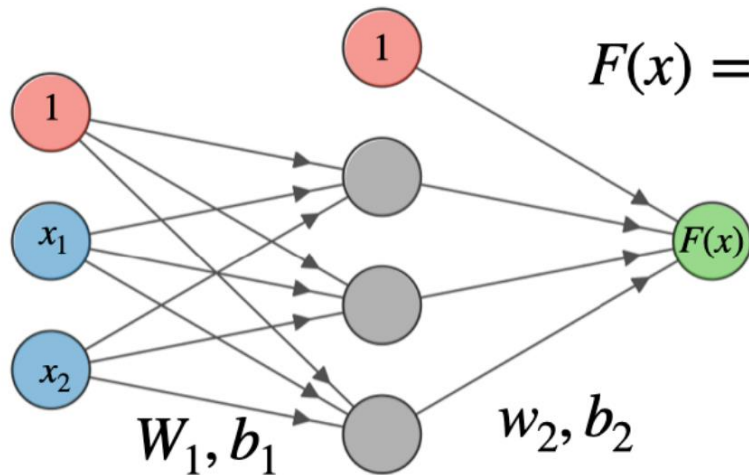
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3. Lets plug this into 1),  $1 = \sum_k \theta_k = \sum_k \frac{n_k}{\lambda} \Rightarrow \lambda = \sum_k n_k = N$ .
4. All together then,  $\theta_k = \frac{n_k}{N}$ .

# Doing MLE requires optimization

$$\theta_{MLE} = \underset{\theta \in \Theta}{\operatorname{argmax}} \log p(D|\theta)$$

- For Gaussians, multinomials, & linear regression, the MLE can be obtained *in closed form* by setting the derivative to zero:  
*e.g.*,  $\theta_k = \frac{n_k}{N}$ .
- **Peek ahead:** what if we had a neural network model?



$$F(x) = w_2^T \sigma(W_1^T x + b_1) + b_2$$

Here, we will need  
*iterative optimization*  
(gradient descent).

Input Layer  $\in \mathbb{R}^3$

Hidden Layer  $\in \mathbb{R}^4$

Output Layer  $\in \mathbb{R}^1$

**“Squared Loss” derived from MLE**

$$L_i((x_i, y_i), \{W_1, w_2, b_1, b_2\}) = (y_i - F(x | W_1, w_2, b_1, b_2))^2$$

# Roadmap

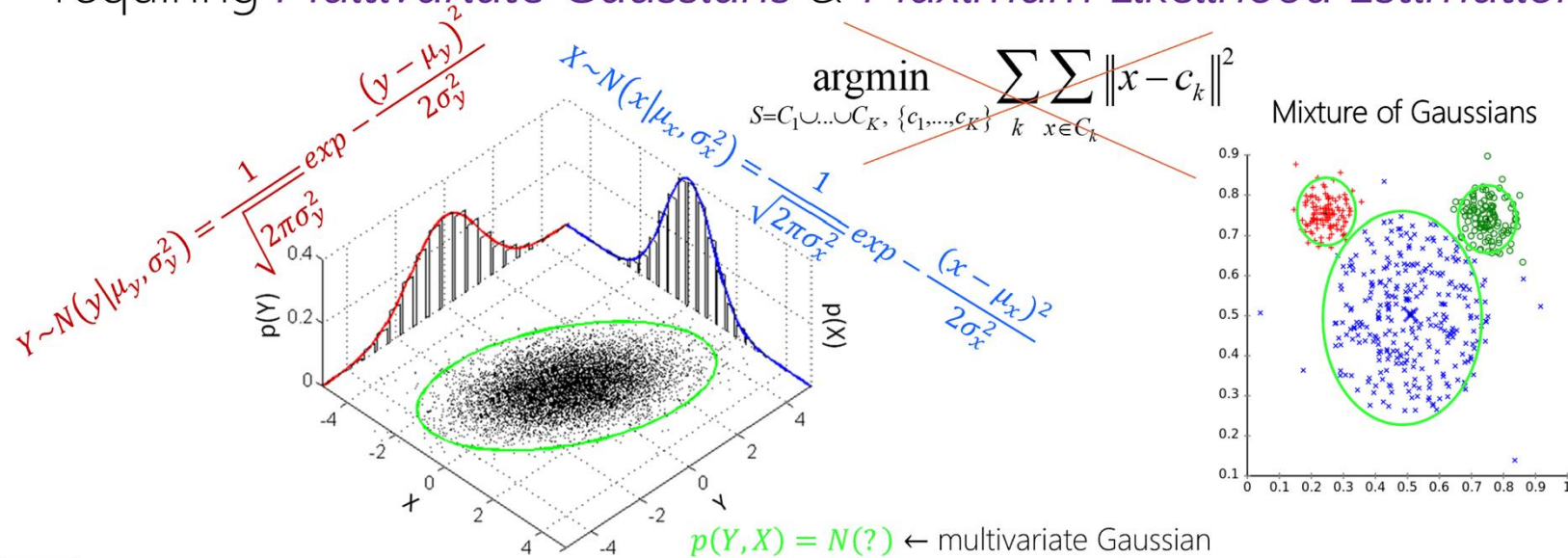
- Finish Maximum Likelihood Estimation (MLE)
- Multivariate Gaussians
- Mixture of Gaussians

# Multivariate Gaussian (MVG) distributions

Recall that the pdf of a univariate Gaussian (normal) distribution is:

$$X \sim \mathcal{N}(x|\mu, \sigma^2) \Rightarrow p(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2\sigma^2}(x - \mu)^2\right)$$

We will need to estimate parameters of a *Mixture of Gaussians*, requiring *Multivariate Gaussians* & *Maximum Likelihood Estimation*.



# Multivariate Gaussian (MVG) distributions

Recall that the pdf of a univariate Gaussian (normal) distribution is:

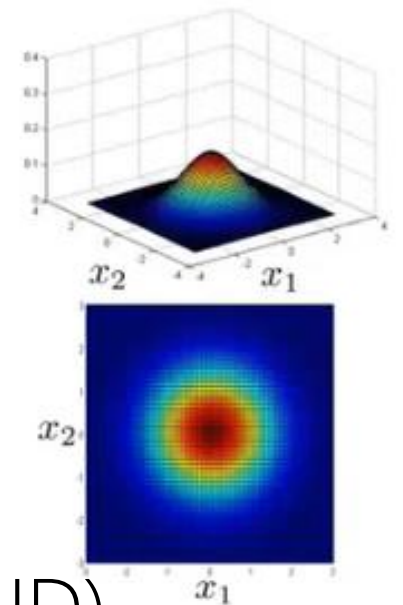
$$X \sim \mathcal{N}(x|\mu, \sigma^2) \Rightarrow p(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(x-\mu)^2\right)$$

*positive semi-definite*

The multivariate extension is:  $x \in \mathbb{R}^d, u \in \mathbb{R}^d$  and  $\Sigma \in \mathbb{R}^{d \times d}$  and PSD.

$$X \sim \mathcal{N}(x|\mu, \Sigma) = \frac{1}{(2\pi)^{D/2}} \frac{1}{|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right)$$

*$\sqrt{|2\pi\Sigma|}$*



MVG also called Multivariate Normal Distribution (MND)

# Carl Friedrich Gauss



Portrait of Gauss by [Christian Albrecht Jensen](#)  
(1840)

## Born

Johann Carl Friedrich Gauss

30 April 1777

[Brunswick, Principality of  
Brunswick-Wolfenbüttel](#)

## Died

23 February 1855 (aged 77)

[Göttingen, Kingdom of  
Hanover, German  
Confederation](#)

# MVGs permeate much of classical and modern day ML:

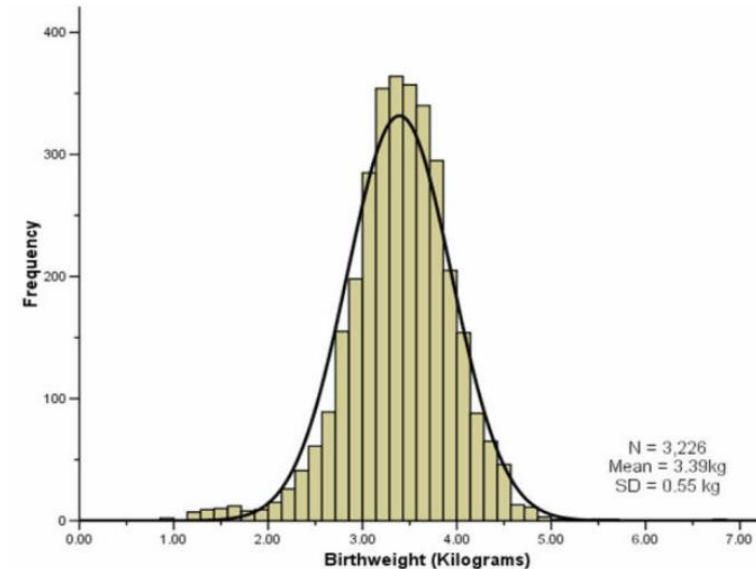
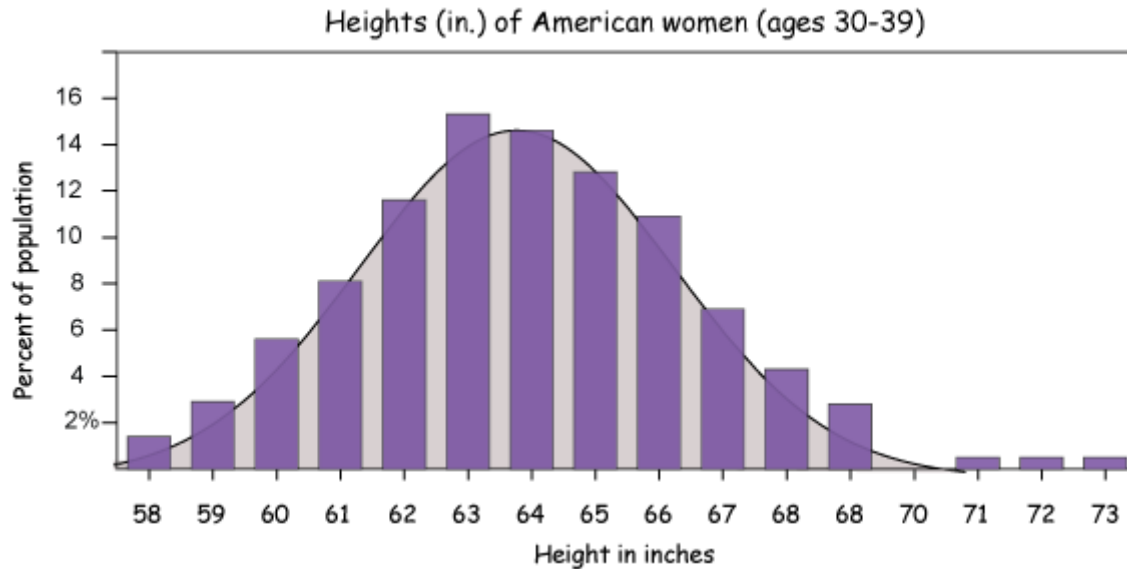
- Classification: generative vs. discriminative.
- Unsupervised models: Principal Components Analysis & autoencoders.
- Advanced topics: Gaussian Process Regression (and deep versions thereof), *etc.*

# Multivariate Gaussian (MVG) distributions

- Consider two quantities, height and weight (of humans).
- Given the arguments of CLT with genetics, it's plausible that each of these is Gaussian distributed, so let's assume:

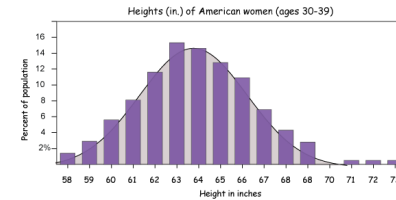
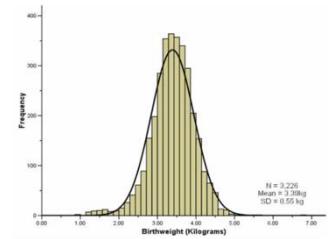
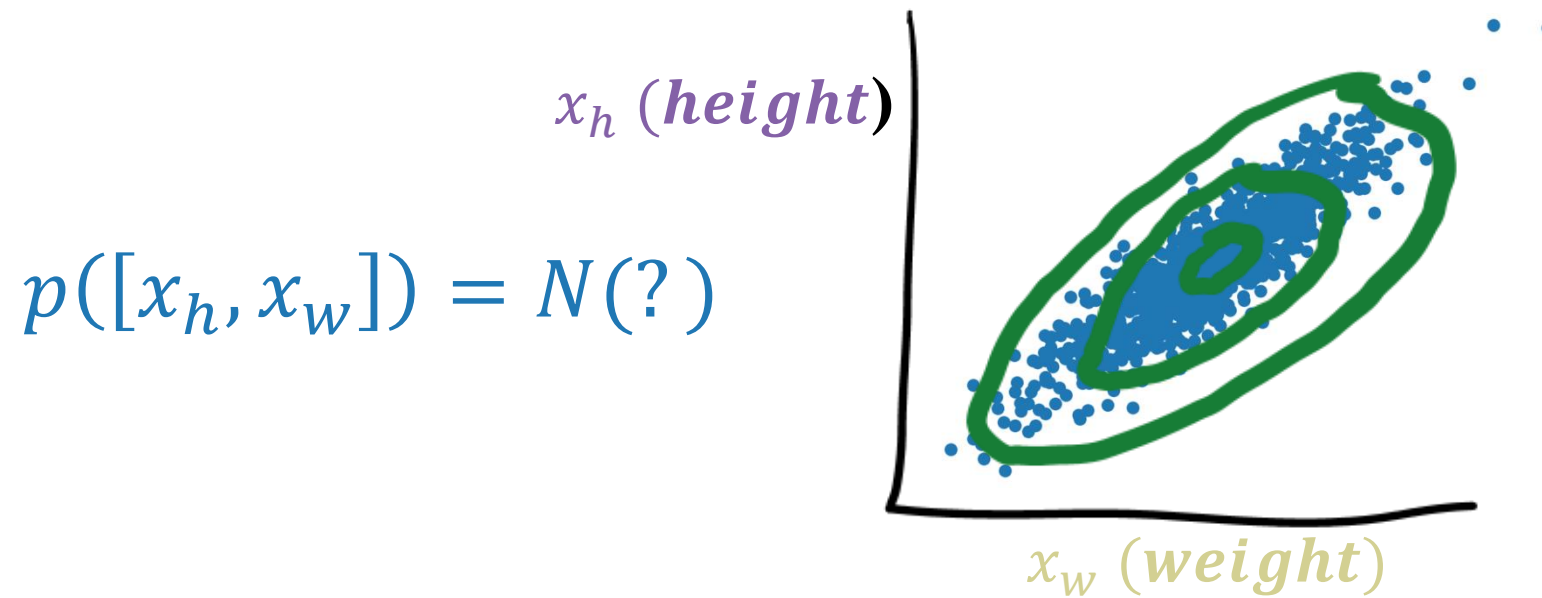
$$\text{height} = X_h \sim N(\mu_h, \sigma_h^2)$$

$$\text{weight} = X_w \sim N(\mu_w, \sigma_w^2)$$



Suppose I want the *joint distribution*,  $p([X_h = x_h, X_w = x_w])$ , how would we write it down? (Shorthand:  $p([x_h, x_w])$ ).

# Multivariate Gaussian (MVG) distributions

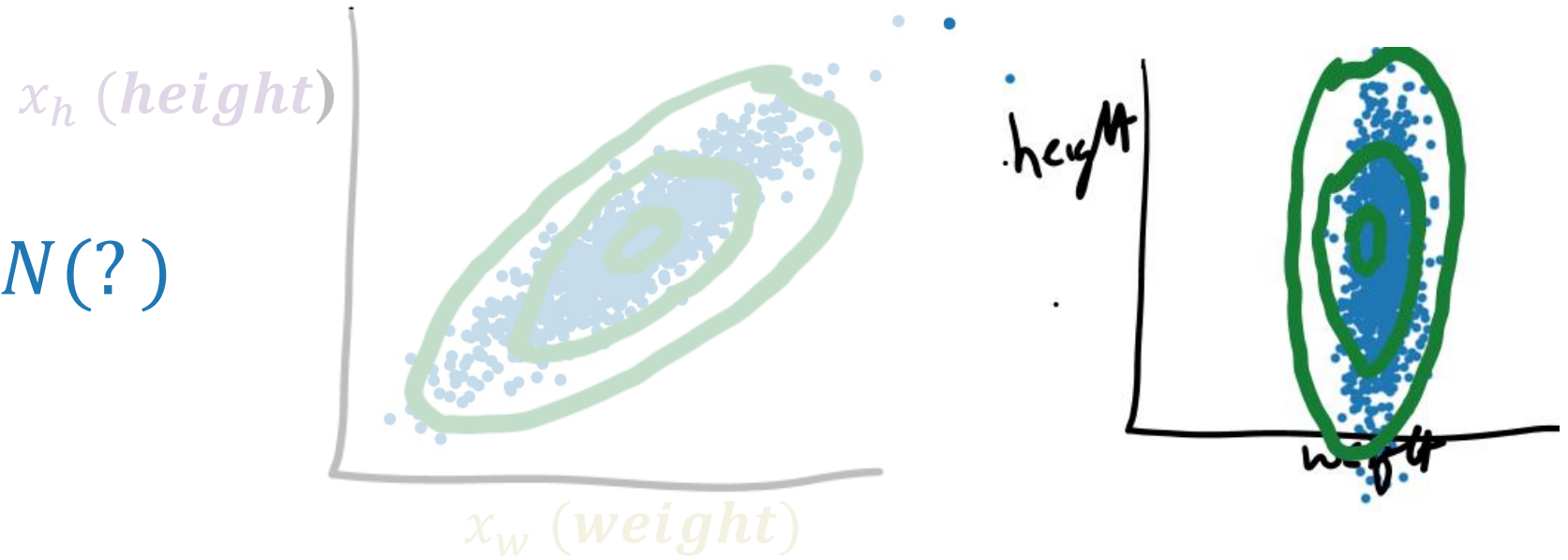


$height = X_h \sim N(\mu_h, \sigma_h^2)$   
 $weight = X_w \sim N(\mu_w, \sigma_w^2)$

- Each point is a sample from some 2D pdf,  $p([x_h, x_w])$ .
- If we computed the mean of this distribution,  $u = [u_1, u_2]$ , it would be..?.
- $u = [\mu_h, \mu_w] = E_{p([x_h, x_w])}[x_h, x_w] = [E[x_h], E[x_w]]$
- How do we describe the “spread” of the points? What dimensionality would it even be?
- Can we use  $p([x_h, x_w]) = N(x_h; \mu_h, \sigma_h^2) * N(x_w; \mu_w, \sigma_w^2)$  ?

# Multivariate Gaussian (MVG) distributions

$$p([x_h, x_w]) = N(?)$$



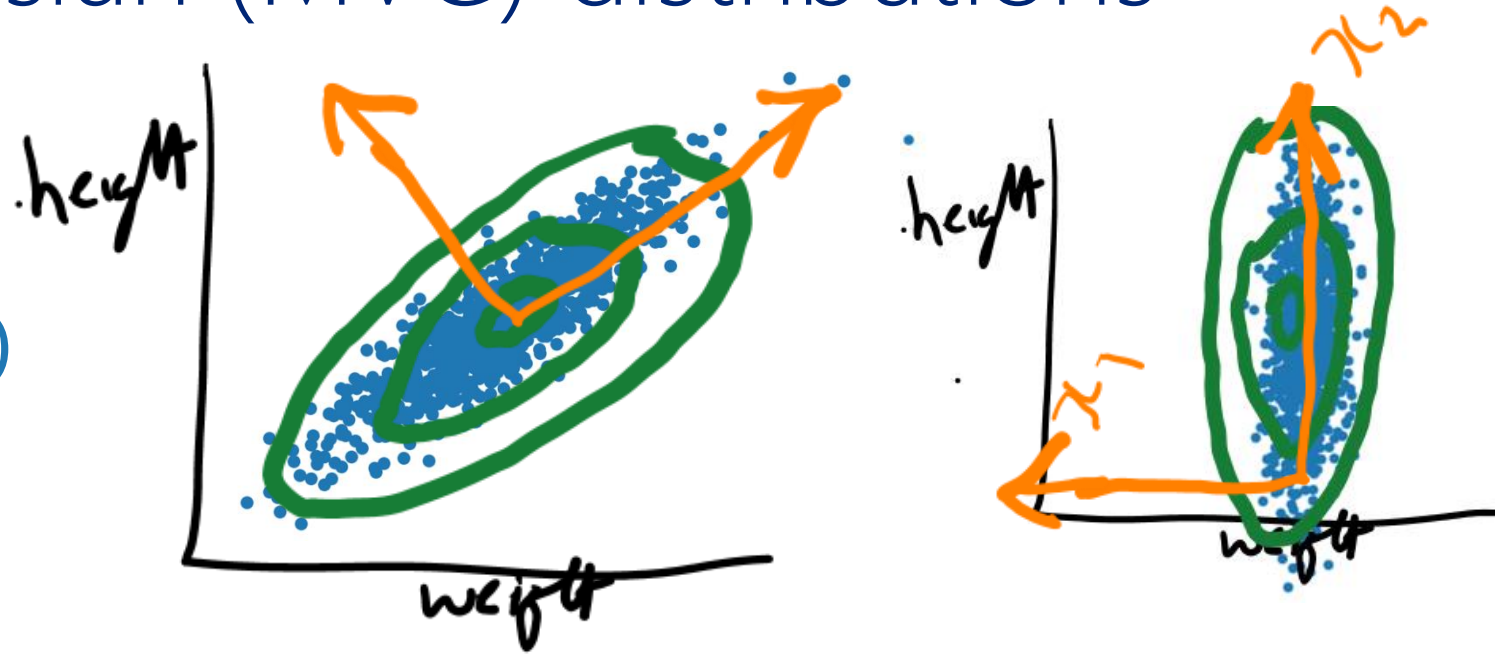
- If *independent* RVs, then  $p([x_h, x_w]) = N(\mu_h, \sigma_h^2) * N(\mu_w, \sigma_w^2)$ .
- If not, then  $p([x_h, x_w]) = N(?)$

$$height = X_h \sim N(\mu_h, \sigma_h^2)$$

$$weight = X_w \sim N(\mu_w, \sigma_w^2)$$

# Multivariate Gaussian (MVG) distributions

$$p([x_h, x_w]) = N(?)$$



- If we could rotate the coordinate system to be “axis aligned”, then

$$p([x_1, x_2]) = N(\mathbf{x}_1; \mu_1, \sigma_1^2) * N(\mathbf{x}_2; \mu_2, \sigma_2^2).$$

- How do we do a rotation?  $Q^T = Q^{-1}$  and  $\det(Q) = 1$
- Multiply by an appropriate orthonormal matrix,  $Q$ :

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = Q \begin{bmatrix} x_h \\ x_w \end{bmatrix}$$

$$height = X_h \sim N(\mu_h, \sigma_h^2)$$

$$weight = X_w \sim N(\mu_w, \sigma_w^2)$$

# Understanding the covariance matrix

Easy case: variables are independent, and each is 1D:

- $X \sim p(x) = N(\mu_1, \sigma_1^2)$  and  $Y \sim p(y) = N(\mu_2, \sigma_2^2)$
- Then  $p([x, y]) = \frac{1}{\sqrt{2\pi\sigma_1^2}} \exp\left[-\frac{1}{2\sigma_1^2}(x - \mu_1)^2\right] \frac{1}{\sqrt{2\pi\sigma_2^2}} \exp\left[-\frac{1}{2\sigma_2^2}(y - \mu_2)^2\right]$

# Understanding the covariance matrix

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$$= \frac{1}{2\pi \sqrt{\sigma_1^2 \sigma_2^2}} \exp\left[-\frac{1}{2\sigma_1^2}(x - \mu_1)^2 - \frac{1}{2\sigma_2^2}(y - \mu_2)^2\right]$$

$$= P(x, y) = \frac{1}{2\pi \sqrt{\sigma_1^2 \sigma_2^2}} \exp\left(-\frac{1}{2} \left\{ \begin{bmatrix} x - \mu_1 & y - \mu_2 \end{bmatrix} \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix} \begin{bmatrix} x - \mu_1 \\ y - \mu_2 \end{bmatrix} \right\}\right)$$

# Understanding the covariance matrix

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math inverse:

$$\begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix}^{-1} = \begin{bmatrix} 1/\sigma_1^2 & 0 \\ 0 & 1/\sigma_2^2 \end{bmatrix}$$

$$\begin{bmatrix} x & y \end{bmatrix} \Sigma^{-1} \begin{bmatrix} x \\ y \end{bmatrix}$$

$\Sigma$  is called the **covariance** matrix in the MVG ( $\Sigma^{-1}$  the precision matrix)

# Understanding the covariance matrix

Easy case: variables are independent, and each is 1D

- $X \sim p(x) = N(\mu_1, \sigma_1^2)$  and  $Y \sim p(y) = N(\mu_2, \sigma_2^2)$

- Then  $p([x, y]) = \frac{1}{\sqrt{2\pi\sigma_1^2}} \exp\left[-\frac{1}{2\sigma_1^2}(x - \mu_1)^2\right] \frac{1}{\sqrt{2\pi\sigma_2^2}} \exp\left[-\frac{1}{2\sigma_2^2}(y - \mu_2)^2\right]$

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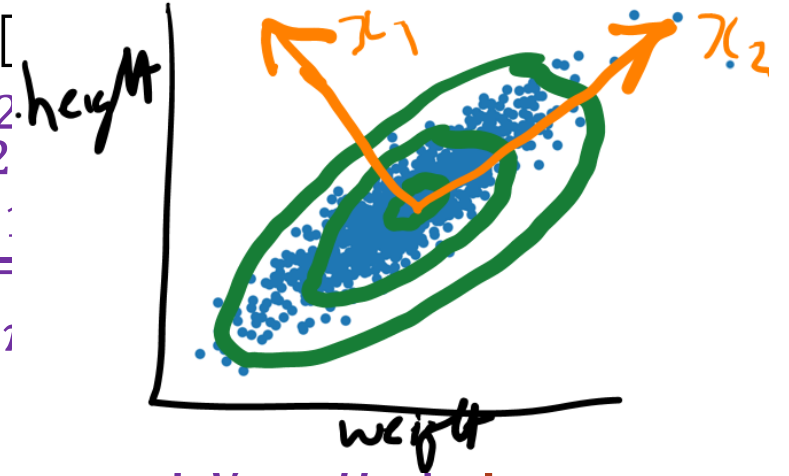
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# Review of expectations, variance, covariance

## EXPECTATION

$E[X]$  for discrete  $\sum x P(x)$   
for continuous  $\int x p(x) dx$   
often denoted by  $\mu$  for mean.

$$E_x[h(x, y)] = \int h(x, y) p(x) dx$$

## Properties of expectation

- Linearity  $E(\sum_i \alpha_i X_i) = \sum_i \alpha_i E(X_i)$  ← doesn't require independence
- Let  $X_1, \dots, X_n$  be independent random variables  
 $E(\prod_{i=1}^n X_i) = \prod_{i=1}^n E(X_i)$
- $E(X+c) = E(X) + c$  ← constant

# Review of expectations, variance, covariance

## VARIANCE

Let  $X$  be a r.v. with mean (expectation)  $\mu = E[X]$

Then variance is defined as  $E(X - \mu)^2$

Properties:

- $V_{\text{ar}}(X) = E(X^2) - \mu^2$
- $\text{Var}(aX + b) = a^2 \text{Var}(X)$
- if  $X_1, \dots, X_n$  are independent and  $\alpha_1, \dots, \alpha_n$  constants  
$$\text{Var}(\sum \alpha_i X_i) = \sum \alpha_i^2 \text{Var}(X_i)$$

# Review of expectations, variance, covariance

COVARIANCE of two R.V.

$$\text{COV}(X, Y) = E((X - E(X))(Y - E(Y))) = E((X - \mu_X)(Y - \mu_Y))$$

$\underbrace{\hspace{10em}}_{\substack{\text{R.V. with} \\ \text{mean 0}}}$

$$= E[XY] - E[X]E[Y]$$

$$\Rightarrow \text{COV}(X, X) = \text{Var}(X) \quad \text{independent } X, Y \Rightarrow \text{COV}(X, Y) = 0$$

CORRELATION

$$\text{CORR}(X, Y) = \frac{\text{COV}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}$$

$$\Rightarrow \rho \in [-1, 1]$$

$\searrow$   
 $r$

# Back to this example

Lets work out the baby case, variables are independent

- $X \sim p(x) = N(\mu_1, \sigma_1^2)$  and  $Y \sim p(y) = N(\mu_2, \sigma_2^2)$

- Then  $p([x, y]) = \frac{1}{\sqrt{2\pi\sigma_1^2}} \exp\left[-\frac{1}{2\sigma_1^2}(x - \mu_1)^2\right] \frac{1}{\sqrt{2\pi\sigma_2^2}} \exp\left[-\frac{1}{2\sigma_2^2}(y - \mu_2)^2\right]$

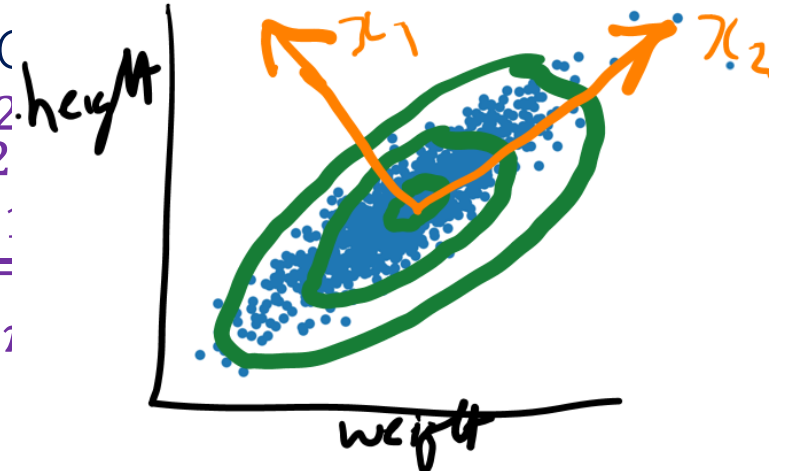
$$= \frac{1}{\sqrt{2\pi\sigma_1^2\sigma_2^2}} \exp\left[-\frac{1}{2\sigma_1^2}(x - \mu_1)^2 - \frac{1}{2\sigma_2^2}(y - \mu_2)^2\right]$$

$$= P(x, y) = \frac{1}{2\pi\sqrt{\sigma_1^2\sigma_2^2}} \exp\left(-\frac{1}{2}\left\{[x - \mu_1, y - \mu_2] \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix}^{-1} \begin{bmatrix} x - \mu_1 \\ y - \mu_2 \end{bmatrix}\right\}\right)$$

lets  
revisit

$$[x \ y] \Sigma^{-1} \begin{bmatrix} x \\ y \end{bmatrix}$$

$\Sigma$  is called the **covariance** matrix in the MVG ( $\Sigma^{-1}$  the precision matrix)



The covariance matrix contains covariances!

$$\begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} \quad \Sigma = \begin{bmatrix} \text{COV}(X_1, X_1) & \text{COV}(X_1, X_2) & \text{COV}(X_1, X_3) \\ \text{COV}(X_2, X_1) & \text{COV}(X_2, X_2) & \text{COV}(X_2, X_3) \\ \text{COV}(X_3, X_1) & \text{COV}(X_3, X_2) & \text{COV}(X_3, X_3) \end{bmatrix}$$

$\nearrow \sigma_1^2$   $\searrow \sigma_3^2$

Symmetric. Since  $\text{COV}(X_1, X_2) = \text{COV}(X_2, X_1)$   
also positive semi definite.

In our example  $\begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix}$  is the Cov. Matrix.  $\text{COV}(X_1, X_2) = 0$  since  
We started from independent variables  $X, Y$ .

# Multivariate Gaussian (MVG) distributions

Fact: If  $X = [X_1, \dots, X_d] \in \mathbb{R}^d$  is distributed as a MVG, then  
 $\forall i, j \in \{1, \dots, d\} \quad \text{cov}(X_i, X_j) = 0$  iff  $X_i, X_j$  are independent.

Generally (beyond MVG), weaker statement: *if*  $X_i, X_j$  are independent *then*  $\text{cov}(X_i, X_j) = 0$ .

Intuition?

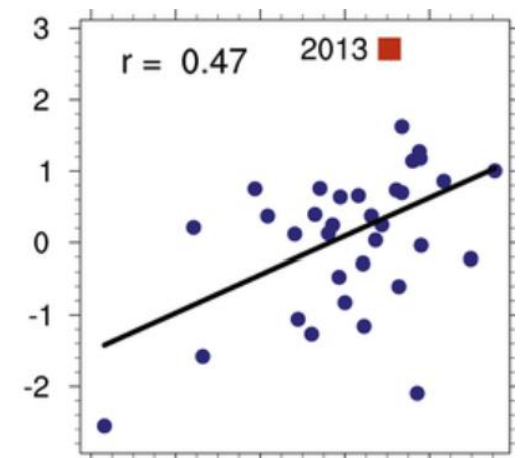
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- Then  $p([x, y]) = \frac{1}{\sqrt{2\pi\sigma_1^2}} \exp\left[-\frac{1}{2\sigma_1^2}(x - \mu_1)^2\right] \frac{1}{\sqrt{2\pi\sigma_2^2}} \exp\left[-\frac{1}{2\sigma_2^2}(y - \mu_2)^2\right]$

$$= \frac{1}{\sqrt{2\pi\sigma_1^2\sigma_2^2}} \exp\left[-\frac{1}{2\sigma_1^2}(x - \mu_1)^2 - \frac{1}{2\sigma_2^2}(y - \mu_2)^2\right]$$

$$= P(x, y) = \frac{1}{2\pi\sqrt{\sigma_1^2\sigma_2^2}} \exp\left(-\frac{1}{2}\begin{bmatrix} x - \mu_1 & y - \mu_2 \end{bmatrix} \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix}^{-1} \begin{bmatrix} x - \mu_1 \\ y - \mu_2 \end{bmatrix}\right)$$

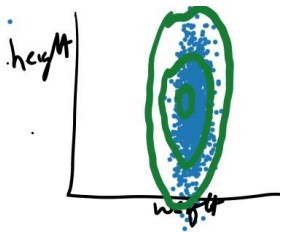


# From the baby case to the general case

We worked the easy case, variables are independent, and each is 1D:

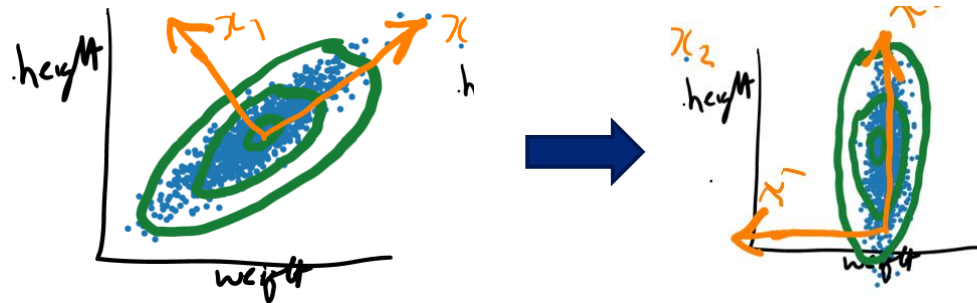
$X \sim p(x) = N(\mu_1, \sigma_1^2)$  and  $Y \sim p(y) = N(\mu_2, \sigma_2^2)$ , so that  $p([x, y]) = p(x)p(y)$

$$P(x, y) = \frac{1}{2\pi \sqrt{\sigma_1^2 \sigma_2^2}} \exp\left(-\frac{1}{2} \left\{ \begin{bmatrix} x - \mu_1 & y - \mu_2 \end{bmatrix} \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix}^{-1} \begin{bmatrix} x - \mu_1 \\ y - \mu_2 \end{bmatrix} \right\}\right)$$



How can we better understand the general case, with  $\mathbf{X} \in \mathbb{R}^d$  and non-independence between the components?

$$p(x; \mu, \Sigma) = \frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu)\right)$$



# The MVG is at its core a quadratic form

MVG has 2 main terms:

1. Quadratic term, where most of the "*action happens*".

$$p(x; \mu, \Sigma) = \frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \exp \left( -\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu) \right)$$

# The MVG is at its core a quadratic form

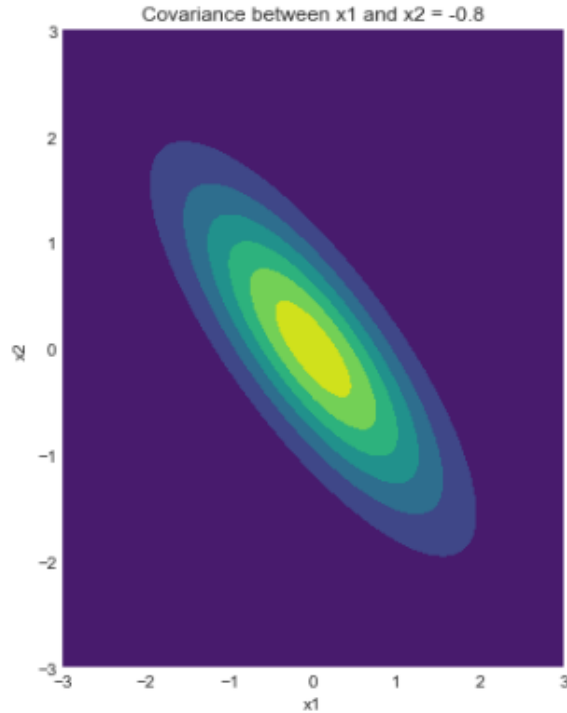
MVG has 2 main terms:

1. Quadratic term, where most of the "*action happens*".
2. Normalizing constant, which ensures that the distribution integrates to 1.

$$p(x; \mu, \Sigma) = \frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \exp \left( -\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu) \right)$$

$$\frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} \exp \left( -\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu) \right) dx_1 dx_2 \cdots dx_n = 1.$$

# Quadratic term $\Rightarrow$ level sets of MVG pdf are ellipses



One level set (contour line) of MVG pdf comprises the values  $x$  for which  $p(x)$  is a constant:

$$\underbrace{x^T}_{1 \times d} \underbrace{\Sigma^{-1}}_{d \times d} \underbrace{x}_{d \times 1} = \text{constant}$$

e.g.  $x \in \mathbb{R}^{d=2}$  then  $0 = ax_1^2 + bx_1x_2 + cx_2^2 + d$

$$p(x; \mu, \Sigma) = \frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \exp \left( -\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu) \right)$$

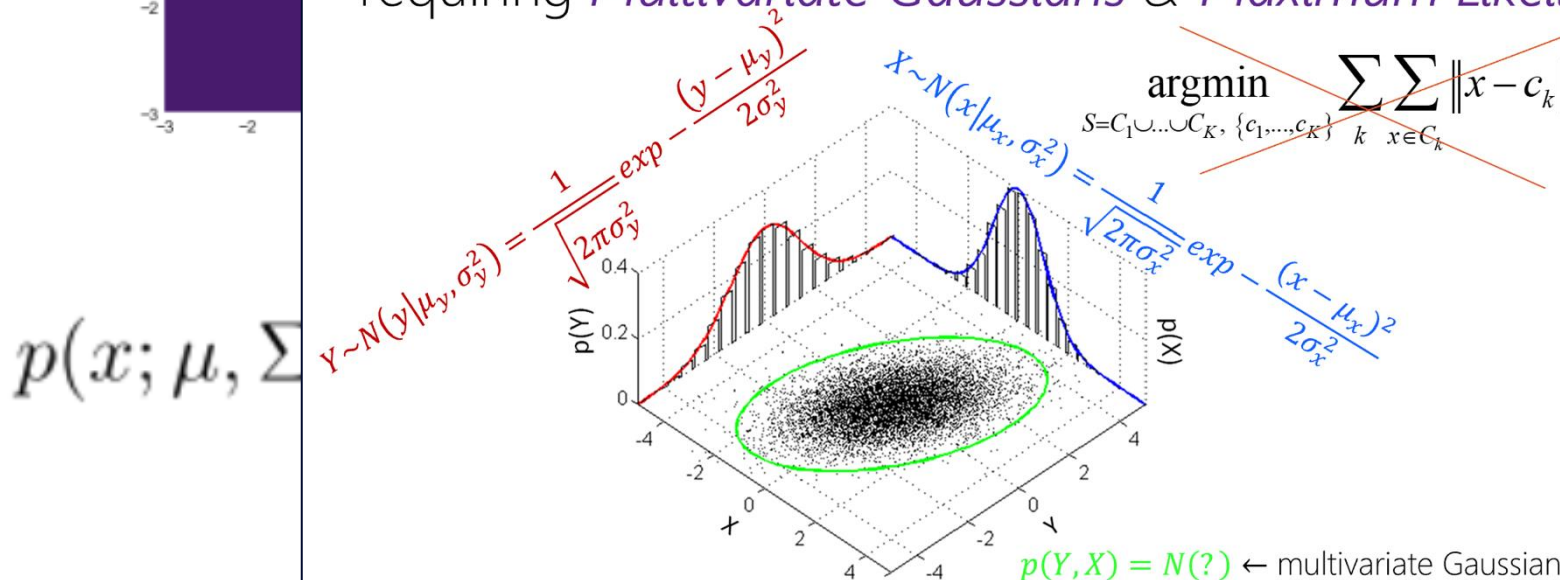
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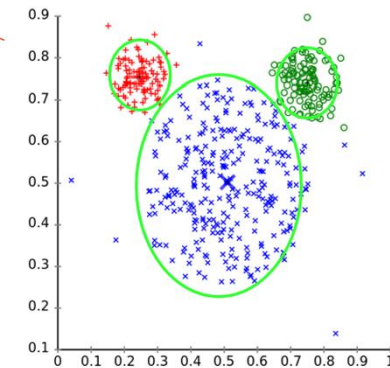
$$x^T \Sigma^{-1} x = \text{constant}$$

We will need to estimate parameters of a *Mixture of Gaussians*, requiring *Multivariate Gaussians* & *Maximum Likelihood Estimation*.



$$\text{argmin}_{S=C_1 \cup \dots \cup C_K, \{c_1, \dots, c_K\}} \sum_k \sum_{x \in C_k} \|x - c_k\|^2$$

Mixture of Gaussians

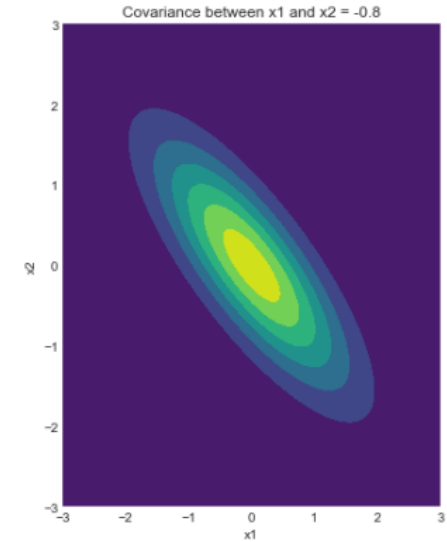


$$x_2^2 + d$$

# What do these look like in high dimensions?

*"To visualize a 14-dimensional space..."*

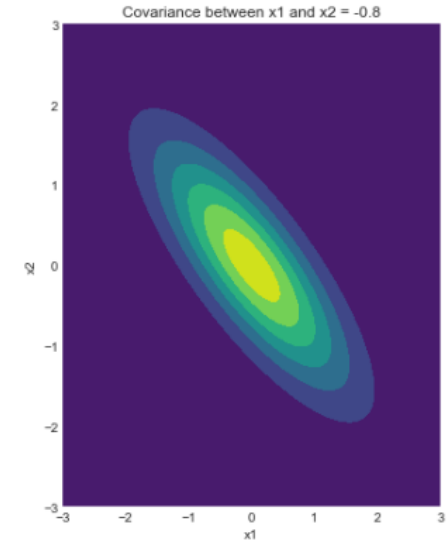
—Geoff Hinton, “grandfather” of deep neural networks (U. Toronto). Nobel Prize 2024.



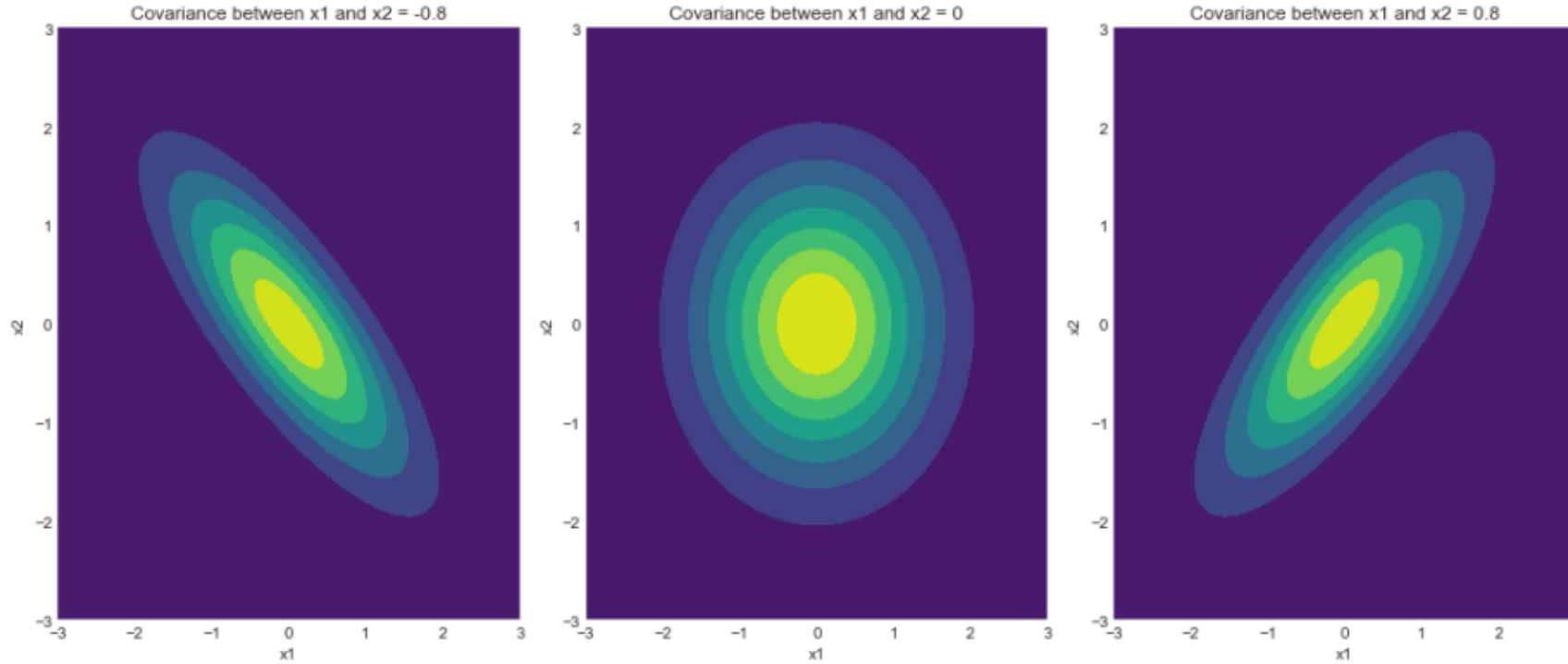
# What do these look like in high dimensions?

*"To visualize a 14-dimensional space... visualize a 3D space and say "fourteen" to yourself very loudly. Everyone does it."*

—Geoff Hinton, "grandfather" of deep neural networks (U. Toronto). Nobel Prize 2024.



# Still, lets try to get an intuition.



could visualize each pair of variables  $x_i$  and  $x_j$

$$p(x; \mu, \Sigma) = \frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \exp \left( -\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu) \right)$$

# Roadmap

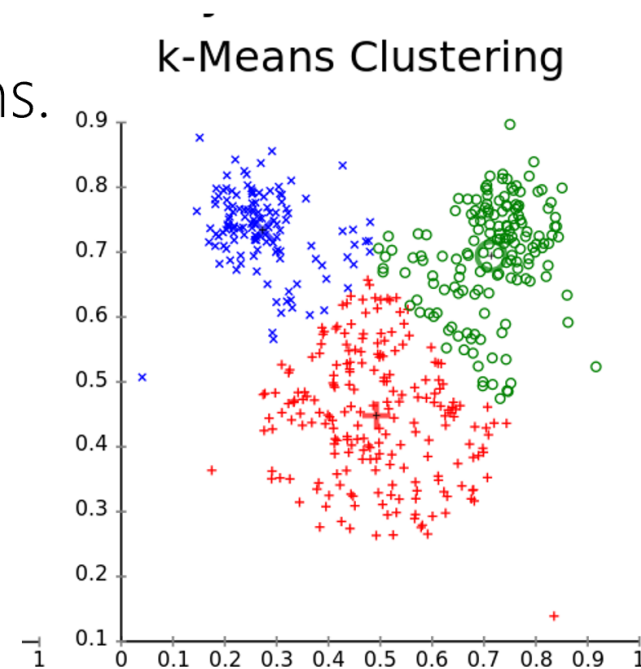
- Finish Maximum Likelihood Estimation (MLE)
- Multivariate Gaussians
- Mixture of Gaussians

# Recall:

Lets revisit K-means—any weaknesses?

$$\operatorname{argmin}_{S=C_1 \cup \dots \cup C_K, \{c_1, \dots, c_K\}} \sum_k \sum_{x \in C_k} \|x - c_k\|^2$$

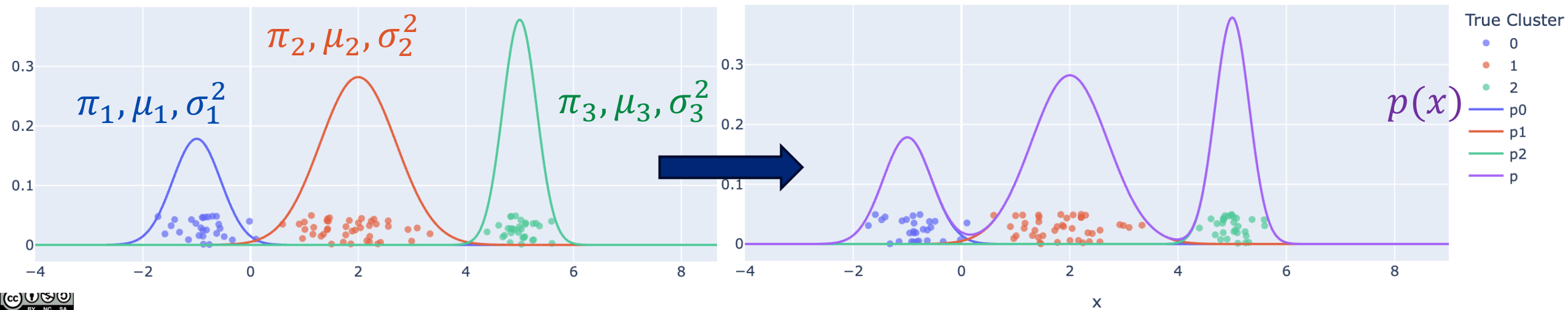
1. No likelihood, so hard to understand assumptions.
2. e.g., implicitly corresponds to clusters with “spherical” shape because each feature is treated equally.
3. Each step in the optimization has a “hard” assignment which means that can’t have any uncertainty as to which point belongs to which cluster, which feels “brittle”



# Gaussian Mixture Model (GMM) *aka* Mixture of Gaussians (MoG)

The Gaussian mixture model,  $p(\mathbf{x})$ , defines the probability of sampling a  $D$ -dimensional vector  $\mathbf{x}$  as a **weighted combination** of Gaussians:

- Example:  $K=3$

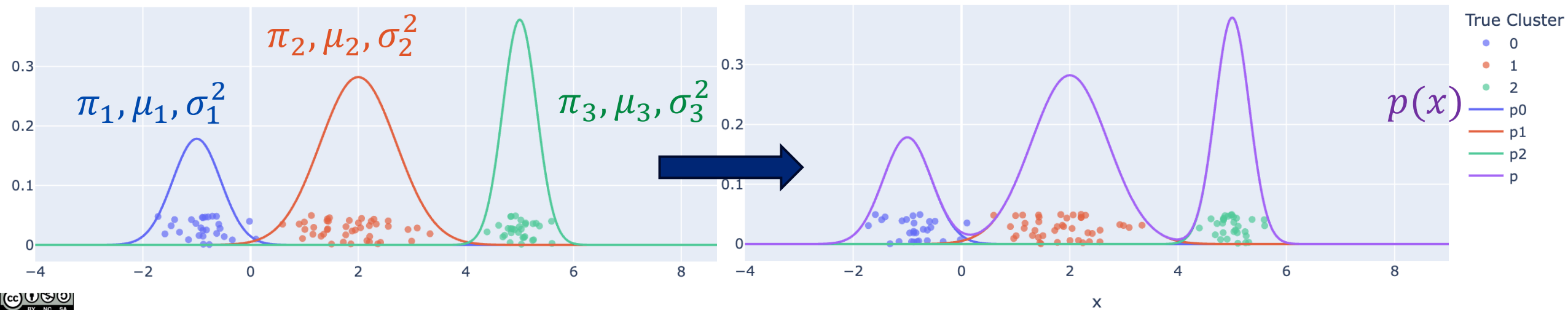


# Mixture of Gaussians

The Gaussian mixture model,  $p(\mathbf{x})$ , defines the probability of sampling a  $D$ -dimensional vector  $\mathbf{x}$  as a **weighted combination** of Gaussians:

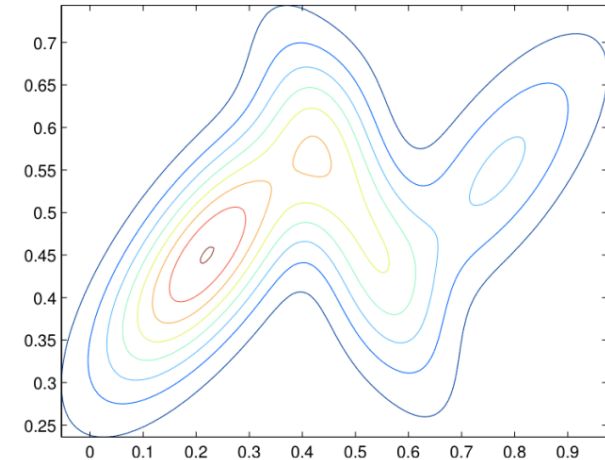
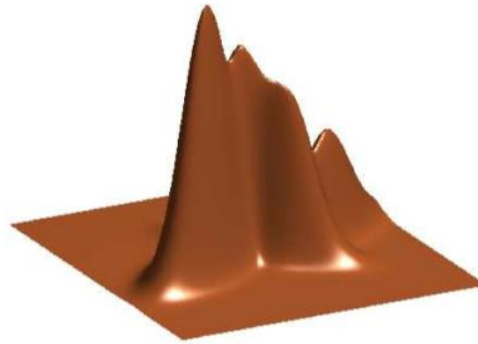
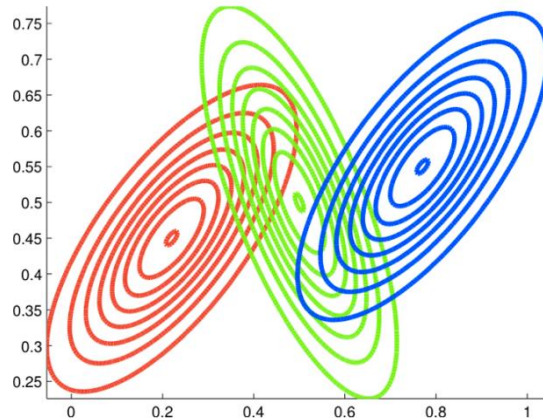
$$p(\mathbf{x} \mid \pi, \mu_1, \Sigma_1, \dots, \mu_K, \Sigma_K) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x} \mid \mu_k, \Sigma_k)$$

where  $\sum_{k=1}^K \pi_k = 1$  is a probability distribution over clusters.



# Mixture of Gaussians (MoG) for clustering

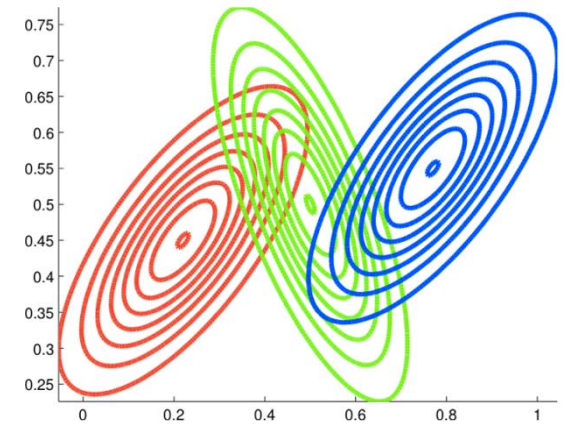
- Each cluster is now represented by a Gaussian  $N(x_i|u_k, \Sigma_k)$ , with two free parameters.
- Now we can write down a **clustering model likelihood** and perform MLE!



# Mixture of Gaussians (MoG) likelihood for one $x_i$

- Let  $z_i \in \{1, \dots, K\}$  be the hidden/unobserved assigned cluster to  $x_i$ . We don't know its value, so have to marginalize it (sum it out):

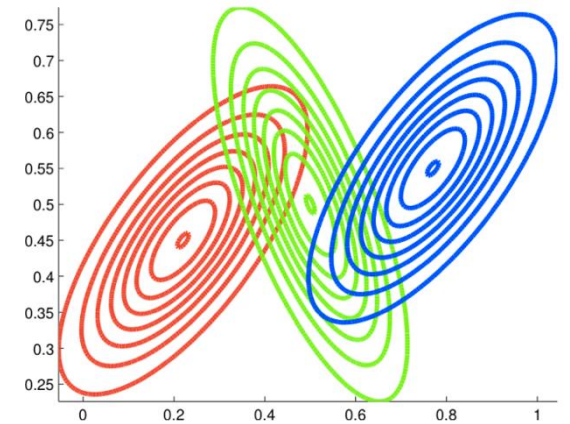
$$L_i = p(x_i | \theta) = \sum_{k=1}^K p(x_i, z_i = k | \theta)$$



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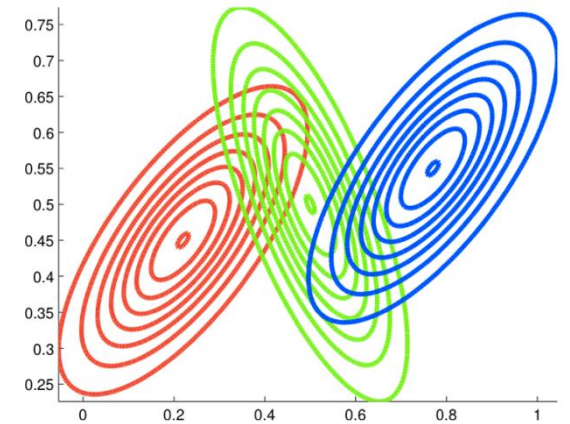
$$\begin{aligned} L_i &= p(x_i | \theta) = \sum_{k=1}^K p(x_i, z_i = k | \theta) \\ &= \sum_{k=1}^K p(x_i | z_i = k, \theta) p(z_i = k | \theta) \end{aligned}$$



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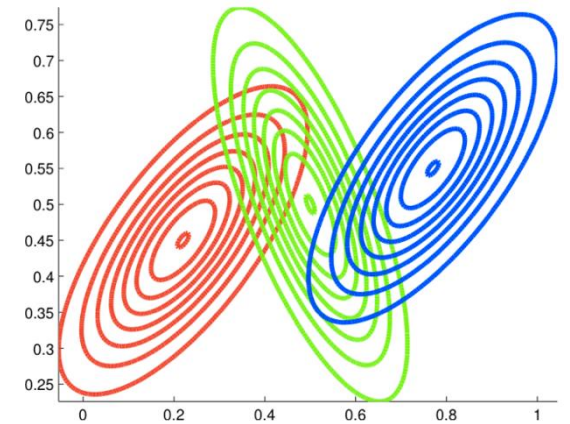


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where  $\alpha_k \equiv p(z_i = k)$  and  $\sum_k \alpha_k = 1$

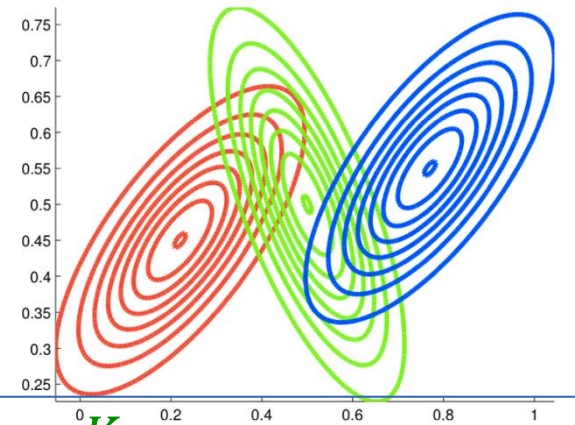


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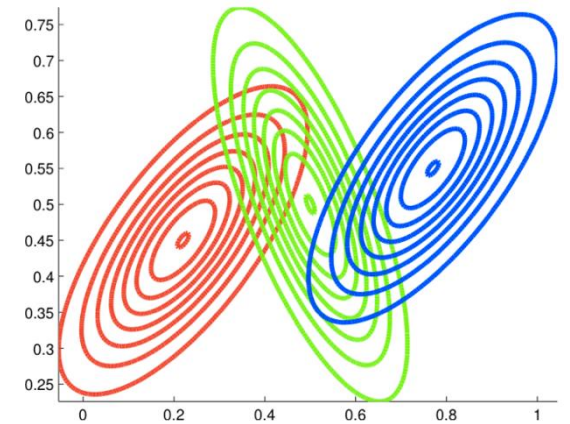


$$LL_i = \log \sum_{k=1}^K N(x_i | \mu_k, \Sigma_k) \alpha_k$$

- The parameters we want to learn are  $\theta \equiv \{\mu_k, \Sigma_k, \alpha_k\}$ .
- $\alpha_k$  are called the "mixing weights".
- Now we can use MLE on  $LL = \log \prod_i L_i = \sum_i \log L_i = \sum_i LL_i$ .

# Mixture of Gaussians (MoG) likelihood for one $x_i$

- But what difficulty will we have?!?!?
- It will not be closed-form (analytical) solution.
- We will need an iterative optimization algorithm: gradient descent, which will get taught later.
- Today we learned how to cast k-means as a probabilistic model.



- The parameters we want to learn are  $\theta \equiv \{\mu_k, \Sigma_k, \alpha_k\}$ .
- $\alpha_k$  are called the "mixing weights".
- MLE:  $\operatorname{argmax}_{\theta} LL = \operatorname{argmax}_{\theta} \log \prod_i L_i = \sum_i \log L_i$ .

# Alternative uses of MoG beyond clustering

Once we have estimated the values of  $\theta = \{\mu_k, \Sigma_k, \alpha_k\}$  from the training data, we can make calls to  $p(x|\theta)$ , for any data point in the training data or otherwise.

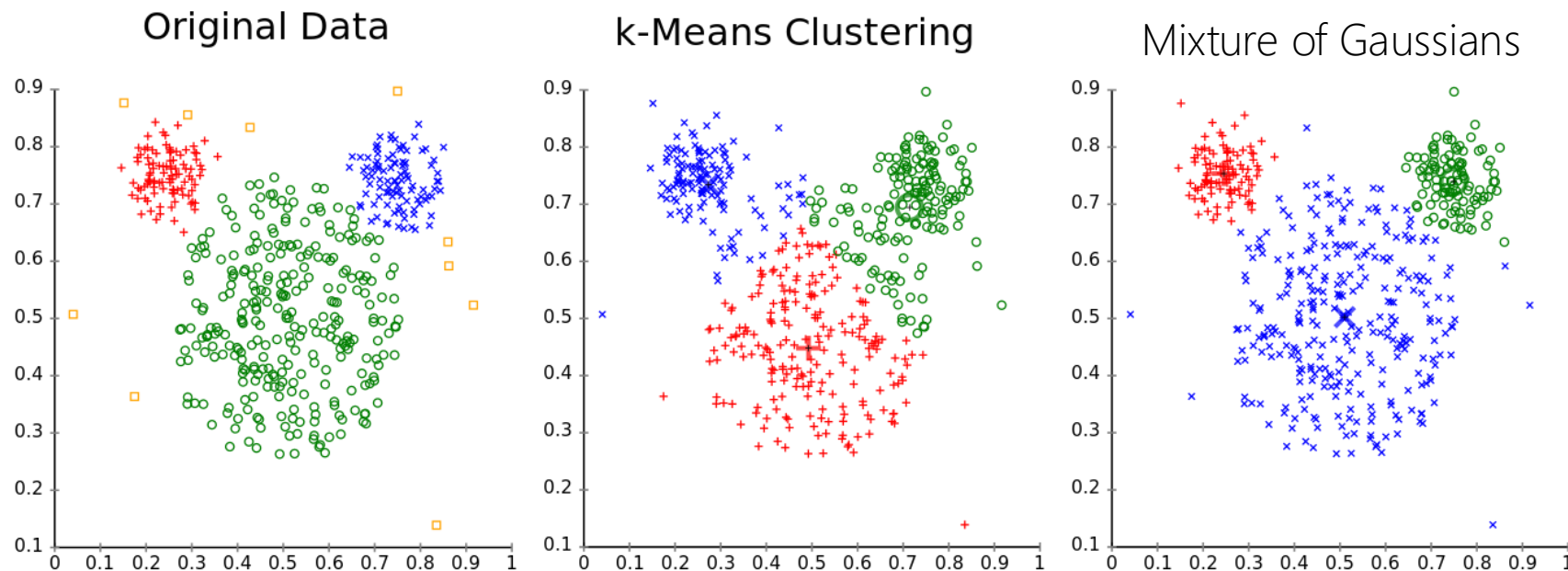
- So we have also performed *density estimation*.

We can also use it to generate data,  $x \sim p(x|\theta)$ .

- So we have a *generative model*:
  1. For each point,  $j$ , sample cluster  $c_j \sim \text{multinomial}(\{\alpha_k\})$ .
  2. Then sample from the corresponding Gaussian.

# K-means vs. Mixture of Gaussians

- If we take the zero noise limit in Mixture of Gaussians (zero variance in the Gaussians), we get K-means.
- Different covariance per cluster is usually helpful, including here.



# K-means vs. Mixture of Gaussians

- MoG: explicit assumptions in the form of statistical distributions.
- Thus easier to generalize, while understanding assumptions.
- Can derive principled objective in the form of a likelihood, which involves marginalizing over the hidden/latent variable (cluster assignment).
- There is a special form of MLE for these latent variables called Expectation-Maximization (see textbook 15.3-15.3.2 if interested)